

Supplementary Material for “Dynamic-Neighborhood Quality-Guided World Models for Online Multi-Car Overtaking in Autonomous Racing”

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TABLE SI
Notation used by DNQ-DLC.

Symbol	Meaning
t	physical decision time
q	arbitrary vehicle index for local geometric definitions
j	surrounding vehicle in an ego-neighbor relation
i	agent-centered row index
k	packed relation-slot index
(n)	candidate rollout index
$t + k t$	k -step prediction made from decision time t
$\hat{x}$	imagined quantity predicted by the world model
$\bar{o}_t$	packed graph observation for the controlled ego row
ϕ_t^i	self feature block of row i
$y_t^{i,k}$	packed relation slot k of row i
$m_t^{i,k}$	validity mask of relation slot k in row i
z_t^i	graph-context embedding of row i
h_t^i	transition hidden state of row i
z_t, h_t	shorthand for z_t^0, h_t^0

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This supplement gives the implementation details, training settings, evaluation protocols, secondary results, and reproducibility information needed to interpret the main paper.

I. Architecture, Packed Observation, and Hyperparameters

A. Notation Summary

B. Packed Observation Layout

The main manuscript denotes the bounded tensorization of the runtime graph by $\bar{o}_t$. The relation capacity is an implementation parameter rather than a fixed physical vehicle topology. In the submitted checkpoint, DNQ-DLC uses

$$d_o = 41 = 17 + 3 \times 8. \quad (1)$$

Thus, the packed observation contains one 17-dimensional self feature block and three masked relation slots:

$$K_{\max} = 3, \quad d_\phi = 17. \quad (2)$$

For the controlled ego row, the k -th relation slot is

$$y_t^{0,k} = [\tilde{r}_{\text{phy},t}^{0,k}, m_t^{0,k}] \in \mathbb{R}^8, \quad (3)$$

where

$$\tilde{r}_{\text{phy},t}^{0,k} = [\Delta f_b, \Delta l_b, \Delta v_x, \Delta v_y, d, \sin \Delta \psi, \cos \Delta \psi]^\top \in \mathbb{R}^7. \quad (4)$$

The complete packed observation is

$$\bar{o}_t = [\phi_t^0, y_t^{0,1}, y_t^{0,2}, y_t^{0,3}] \in \mathbb{R}^{41}. \quad (5)$$

The relation encoder consumes only the seven physical components $\tilde{r}_{\text{phy},t}^{0,k}$; the mask $m_t^{0,k}$ is used for padding control and masked aggregation. In the submitted telemetry builder, Δf_b and Δl_b are ego-body forward and left displacements normalized by PLAYFIELD, whereas Δv_x and Δv_y are global velocity differences normalized by 50. Therefore, the relation slot is an ego-conditioned telemetry relation rather than a purely centerline-frame relation.

C. Validity Mask Generation

The validity mask is generated during bounded graph tensorization and is not learned. Let $\mathcal{N}_t$ be the eligible interaction neighborhood constructed by the runtime selector, and let $j_t^k = \text{TopK}_k(\mathcal{N}_t; \rho)$ be the k -th highest-relevance neighbor when it exists. For the controlled ego row,

$$m_t^{0,k} = \mathbb{I}[k \leq \min(|\mathcal{N}_t|, K_{\max})], \quad k = 1, \dots, K_{\max}. \quad (6)$$

The packed physical relation is

$$\tilde{r}_{\text{phy},t}^{0,k} = \begin{cases} r_t^{0j_t^k}, & m_t^{0,k} = 1, \\ \mathbf{0}, & m_t^{0,k} = 0. \end{cases} \quad (7)$$

Thus, padded slots contain zero physical relation features and have zero validity masks. When $|\mathcal{N}_t| > K_{\max}$, all vehicles are scored by the interaction selector, but only the $K_{\max}$ highest ranked vehicles are tensorized into the bounded masked representation used at that decision step. The physical graph and represented identities are nevertheless rebuilt from telemetry at every real step.

The submitted checkpoint uses $K_{\max} = 3$ as a deliberate capacity-latency trade-off. This setting yielded usable performance in the evaluated scenarios, but its causal sufficiency was not isolated. More complex congestion may require a larger slot capacity or hierarchical selection. Sensitivity to $K_{\max}$ is left as future work.

D. Self Feature Block

The self feature block $\phi_t^i \in \mathbb{R}^{17}$ is the normalized row-wise vehicle and local-map descriptor exported by the telemetry extractor. It is fixed across training and evaluation. It combines global state, track-relative

TABLE SII

Self feature block ϕ_t^i used in the submitted checkpoint. All continuous entries are normalized before being passed to the network. The order is taken from the telemetry-dynamic observation builder used for the submitted checkpoint.

Index	Symbol / Code field	Meaning	Normalization
1	ego_x	global x position of vehicle i	$p_x^i/\text{PLAYFIELD}$
2	ego_y	global y position of vehicle i	$p_y^i/\text{PLAYFIELD}$
3	ego_vx	global x velocity	$v_x^i/50$
4	ego_vy	global y velocity	$v_y^i/50$
5	ego_speed	speed magnitude	$\ v^i\ _2/50$
6	ego_heading_sin	sine of vehicle heading	$\sin \psi^i$
7	ego_heading_cos	cosine of vehicle heading	$\cos \psi^i$
8	ego_angular_velocity	yaw angular velocity	$\dot{\psi}^i/5$
9	track_progress	nearest centerline-index progress	$\eta^i/\max(\mathcal{T} - 1, 1)$
10	tile_progress	visited-tile lap progress	$n_{\text{tile}}^i/ \mathcal{T} $
11	centerline_dx	global x offset from nearest centerline point	$(p_x^i - c_x(\eta^i))/\text{TRACK_WIDTH}$
12	centerline_dy	global y offset from nearest centerline point	$(p_y^i - c_y(\eta^i))/\text{TRACK_WIDTH}$
13	lateral_error	signed centerline-normal offset	$\langle p^i - c(\eta^i), \hat{\mathbf{n}}(\eta^i) \rangle / \text{TRACK_WIDTH}$
14	heading_error_sin	sine of centerline-heading error	$\sin(\psi_c(\eta^i) - \psi^i)$
15	heading_error_cos	cosine of centerline-heading error	$\cos(\psi_c(\eta^i) - \psi^i)$
16	on_grass	off-track exposure flag	binary
17	driving_backward	reverse-driving flag	binary

progress, centerline-offset features, heading-error features, and driving-status flags.

Because the submitted self feature block includes normalized global position and velocity fields, the representation should be interpreted as map-conditioned rather than strictly map-relative in every channel. The track-relative components provide local map context, while the global components preserve compatibility with the telemetry builder used for the submitted checkpoint.

E. Network Modules

F. Runtime Interaction Selector

The runtime interaction selector uses the implemented interaction mode rather than a direct numeric instantiation of the symbolic $\text{Score}_{\text{int}}(\cdot)$ in the main manuscript. The local radius is $R_{\text{loc}} = 80$ m. A vehicle is treated as forward traffic when

$$0 < \Delta f_b < 70 \text{ m}, \quad |\Delta l_b| < 24 \text{ m},$$

as alongside traffic when

$$|\Delta f_b| < 14 \text{ m}, \quad |\Delta l_b| < 18 \text{ m},$$

and as rear pressure when

$$-18 \text{ m} < \Delta f_b \leq 0, \quad |\Delta l_b| < 14 \text{ m},$$

and closing speed exceeds 1 m/s. The implemented score uses weights 3.6 for forward occupancy, 2.8 for alongside overlap, 1.7 for rear pressure, 1.1 for inverse distance gap, 0.25 for closing speed, 0.18 for heading alignment, and -0.015 for metric distance. There is no separate positive ρ_{min} in the submitted runtime code; vehicles that fail the radius or interaction gate receive a large negative score and are excluded before top-slot packing.

For online ego planning, non-controlled agent-centered row actions are filled by background telemetry policies. At each imagined step, the implementation first evaluates the configured background policy cycle, telemetry_cruise,

telemetry_yield, and telemetry_lane, for non-target vehicles. The target row is then overwritten by the committed candidate action at $k = 0$ or by the blended base/quality actor continuation for $k > 0$. Thus, a_t^i for $i \neq 0$ is generated by the runtime background-policy model during online planning.

G. Numerical Configuration

II. Labels, Losses, and Training

A. World-Model and Quality-Actor Data

The submitted world model was trained on 160 simulator episodes, 347963 transitions, and 1775508 agent samples collected with 4–6 vehicles using telemetry lane, yield, cruise, and overtaking expert policies. The optimizer was Adam with learning rate 3×10^{-4} , batch size 512, 1800 gradient steps, and gradient clipping at 25.0. The original training run did not reserve an episode-disjoint validation split. To avoid reporting training loss as predictive validity, the revised evidence package evaluates the frozen checkpoint on new seeds 7300–7307 for procedural $N = 4$ –6, 7400–7407 for Hairpin $N = 8$, and 7500–7507 for S-Curve $N = 8$. Each stratum contains 4800 transitions. One-step state/reward errors, boundary-risk AUROC, AUPRC, Brier score, 10-bin expected calibration error, and $H \in \{1, 3, 5, 10\}$ rollout errors are stored in heldout_world_model_validation.json. The validation data were generated after the checkpoint was frozen and were not used for model selection.

The boundary-risk label for agent i at $t + 1$ is

$$\chi_{t+1}^i = \mathbb{I}[o_{t+1}^{i,\text{off}} = 1 \vee o_{t+1}^{i,\text{back}} = 1 \vee |\ell_{t+1}^i| > 0.45]. \quad (8)$$

This label captures boundary and unstable-driving risk. Close-gap proximity is handled separately by C_{close} in the planner.

The implementation includes an auxiliary quality target

$$\mathbf{q}_{t+1}^{i,\text{qual}} = [q_{\text{track}}, q_{\text{off}}, q_{\text{lat}}, q_{\text{clear}}]^\top. \quad (9)$$

TABLE SIII
Concrete learned network architecture used by DNQ-DLC. Hidden layers use ELU activation.

Module	Architecture	Input $\rightarrow$ Output	Role
MLP _{ego}	Linear(17, d_h)-ELU-Linear(d_h , d_h)-ELU-Linear(d_h , d_h)	$\phi_t^i \rightarrow e_{\text{self},t}^i$	Self encoding
MLP _{rel}	Linear(7, d_h)-ELU-Linear(d_h , d_h)-ELU-Linear(d_h , d_h)	$\tilde{r}_{\text{phy},t}^{i,k} \rightarrow e_{\text{rel},t}^{i,k}$	Relation-slot encoding
Linear _{att}	Linear(d_h , 1) followed by masked softmax or masked mean	$\{e_{\text{rel},t}^{i,k}, m_t^{i,k}\} \rightarrow \bar{e}_{\text{rel},t}^i$	Masked aggregation
MLP _{act}	Linear(3, d_h)-ELU-Linear(d_h , d_h)-ELU-Linear(d_h , d_h)	$a_t^i \rightarrow e_{\text{act},t}^i$	Action encoding
MLP _{fusion}	Linear(5 d_h , d_h)-ELU-Linear(d_h , d_h)-ELU-Linear(d_h , d_h)	$[z_t^i, e_{\text{act},t}^i, \bar{e}_{\text{self},t}, \bar{e}_{\text{act},t}] \rightarrow h_t^i$	Transition representation
Prior head	Diagonal Gaussian linear heads	$h_t \rightarrow p_\theta(s_t h_t)$	DLC-style latent interface
Posterior head	Diagonal Gaussian linear heads	$(h_t, \bar{o}_{t+1}) \rightarrow q_\phi(s_t h_t, \bar{o}_{t+1})$	Training posterior
Linear _{obs}	Linear heads from $\xi_t = [h_t, s_t]$	$\xi_t \rightarrow (\mu^o, \lambda^o)$	Observation prediction
Linear _{rew}	Linear heads from ξ_t	$\xi_t \rightarrow (\mu^r, \lambda^r)$	Reward prediction
Linear _{risk}	Linear head followed by sigmoid	$\xi_t \rightarrow z^\chi \rightarrow \hat{\chi}$	Boundary-risk prediction
Linear _{qual}	Linear head	$\xi_t \rightarrow \hat{\mathbf{q}}^{\text{qual}}$	Inactive auxiliary head in submitted checkpoint
ActorMLP _{base}	Self MLP $17 \rightarrow d_p \rightarrow d_p \rightarrow d_p$; relation MLP $7 \rightarrow d_p \rightarrow d_p \rightarrow d_p$; actor MLP $2d_p \rightarrow d_p \rightarrow d_p \rightarrow 3$	$\bar{o}_t \rightarrow a_t^{\text{base}}$	Base proposal
ActorMLP _{qual}	Same architecture as base actor	$\bar{o}_t \rightarrow a_t^{\text{qual}}$	Quality-biased proposal

TABLE SIV
Frozen numerical configuration used in the submitted DNQ-DLC evaluation.

Quantity	Symbol	Value
Graph observation dimension	d_o	41
Self feature dimension	d_ϕ	17
Relation slot count	K_{max}	3
Action dimension	d_a	3
World-model hidden dimension	d_h	192
Actor hidden dimension	d_p	128
Quality head dimension	–	4
Rollout horizon	H	4
Candidate budget	–	12
Risk-loss weight	λ_χ	1.20
Quality auxiliary weight	λ_q	0.00
Latent regularization	β_{KL}	0.00
Discount factor	γ	0.985
Progress weight	λ_{prog}	1.20
Overtaking weight	λ_{ovt}	2.40
Off-track penalty	λ_{off}	2.00
Lateral-quality penalty	λ_{lat}	0.42
Close-gap penalty	λ_{close}	0.65
Risk penalty	λ_{risk}	1.65
Uncertainty penalty	λ_{unc}	0.40
Actor imitation penalty	λ_{imit}	0.06
Actor rollout blend	α_q	0.25
Planner target speed	v_{ref}	22.0 m/s

The submitted main checkpoint has $\lambda_q = 0$, so these quality-head targets were implemented but did not contribute to its training objective. Quality-guided action proposals instead come from a separate Graph-BC check-

point trained on 1526 selected states from 24 episodes (seeds 5200–5223). The revised validation uses disjoint seeds 8200–8207, 8300–8307, and 8400–8407 for the procedural, Hairpin, and S-Curve strata, respectively. This test measures conditional action imitation on states selected around accepted overtaking events; it is not a closed-loop safety metric.

B. Baseline Training Budgets

Rule Expert and Safety Rule are author-implemented deterministic telemetry controllers rather than algorithms adopted from an external paper. Rule Expert gates among lane-following, adaptive, overtaking, and recovery behaviors using local telemetry. Safety Rule uses the barrier-expert implementation, which adds density-dependent speed reduction and stricter lateral and heading recovery thresholds. Their registry names are `rule_expert_gate` and `rule_safety_gate`, respectively.

PPO, SAC, and TD3 follow their original algorithm definitions [1]–[3] and are implemented with Stable-Baselines3 [4]. For the RL training-budget sensitivity study, they are trained from three independent seeds (9101–9103), each for 200000 environment steps with four vehicles. On every reset, the simulator seed is incremented, the procedural track is regenerated, background start order is shuffled while the target remains last, line spacing is sampled from {4,5,6}, and lateral spacing is jittered. Checkpoints are

stored in seed-specific directories, preventing the overwrite bug in the earlier training script. Wall-clock training time is approximately 2.0 h per PPO seed, 2.4 h per SAC seed, and 2.2 h per TD3 seed on the reported workstation. Frozen checkpoints are evaluated on seeds 9701–9712 with the same 600-step horizon used for every controller. The source report first aggregates 12 evaluation cases within each training seed and then reports the mean and range over three training seeds. The range is not a confidence interval. No best-of-seed checkpoint selection is performed. Because the training run did not store dense learning curves or intermediate checkpoints, convergence diagnostics remain a limitation despite the larger budget.

DLC-IT, DLC-JT, and DLC-JTO are our implementations of the individual-transition, joint-transition, and joint-transition-plus-observer variants in Deep Latent Competition [5]. Each variant was trained from seeds 0–3 using 120 simulator episodes, 96000 transitions, and 80 actor/world-model updates. Checkpoints were retained after 20, 40, and 80 updates. Seed 1 was selected by the predefined mean score from within-suite round-robin validation. The DLC training budget differs from DNQ-DLC and does not reproduce the original 500-race protocol exactly.

C. Targets, Losses, and Submitted Checkpoint

The four implemented components of $\mathbf{q}^{\text{qual}}$ are

$$q_{\text{track}} = \mathbb{I}[o^{\text{off}} = 0, o^{\text{back}} = 0, |\ell| \leq 0.46, \cos e_\psi \geq 0.70], \quad (10)$$

$$q_{\text{off}} = \mathbb{I}[o^{\text{off}} = 1], \quad (11)$$

$$q_{\text{lat}} = \frac{\text{clip}(|\ell| + 0.55|e_\psi|, 0, 2.2)}{2.2}, \quad (12)$$

$$q_{\text{clear}} = \begin{cases} \min_{j \in \mathcal{A}_{\text{front}}} \text{clip}\left(\frac{\Delta f_{0j}}{65}, 0, 1\right), & |\mathcal{A}_{\text{front}}| > 0, \\ 1, & |\mathcal{A}_{\text{front}}| = 0. \end{cases} \quad (13)$$

Here $\mathcal{A}_{\text{front}}$ contains valid vehicles ahead within 65 m and with lateral separation below 18 m. Larger values are better for q_{track} and q_{clear} , while smaller values are better for q_{off} and q_{lat} .

The next-observation distribution is

$$p_\theta(\bar{o}_{t+1}|h_t, s_t) = \mathcal{N}(\mu_{t+1|t}^o, \text{diag}(\exp(\lambda_{t+1|t}^o))), \quad (14)$$

and the reward distribution is

$$p_\theta(r_{t+1}|h_t, s_t) = \mathcal{N}(\mu_{t+1|t}^r, \exp(\lambda_{t+1|t}^r)). \quad (15)$$

The observation negative log-likelihood is

$$\mathcal{L}_o = \frac{1}{2} \sum_{d=1}^{d_o} [\exp(-\lambda_{t+1|t,d}^o) (\bar{o}_{t+1,d} - \mu_{t+1|t,d}^o)^2 + \lambda_{t+1|t,d}^o], \quad (16)$$

and the reward negative log-likelihood is

$$\mathcal{L}_r = \frac{1}{2} [\exp(-\lambda_{t+1|t}^r) (r_{t+1} - \mu_{t+1|t}^r)^2 + \lambda_{t+1|t}^r]. \quad (17)$$

The auxiliary quality loss is

$$\mathcal{L}_q = \|\hat{\mathbf{q}}_{t+1|t}^{\text{qual}} - \mathbf{q}_{t+1}^{\text{qual}}\|_2^2. \quad (18)$$

The general loss is

$$\mathcal{L}_{\text{wm}} = \mathbb{E}_{q_\phi} [\mathcal{L}_o + \mathcal{L}_r + \lambda_\chi \text{BCE}(\chi_{t+1}, \hat{\chi}_{t+1|t}) + \lambda_q \mathcal{L}_q] + \beta_{\text{KL}} D_{\text{KL}}(q_\phi \| p_\theta). \quad (19)$$

In the submitted checkpoint, $\lambda_q = 0.00$ and $\beta_{\text{KL}} = 0.00$. The auxiliary quality loss and KL term are therefore inactive. Quality guidance comes from the separately trained quality actor, candidate actions, and multi-objective scoring. Online planning uses deterministic mean rollouts, with uncertainty computed from predicted observation and reward log variances.

D. Actor Training

The quality actor was trained by behavior cloning on a filtered overtaking dataset. The dataset contains 24 collection episodes and 1526 state-action samples. Samples were retained from on-track overtaking events using a window from 30 steps before the event to 45 steps after completion. The accepted target policies were telemetry expert gate, fast expert, barrier expert, and adaptive expert policies. The stored filtering mode is `on_track`; failed events are not retained.

The base actor is trained jointly with the Graph-DLC transition model. It uses the same collected world-model dataset described above. During training, the proposal loss is the mean-squared error between $\text{ActorMLP}_{\text{base}}(\bar{o}_t)$ and the collected normalized action. The transition model and base actor share one Adam optimizer with learning rate 3×10^{-4} , batch size 512, 1800 gradient steps, proposal-loss weight 0.35, and gradient clipping at 25.0.

III. Candidate Generation and Scoring

The submitted evaluation scores 12 candidate actions per decision step when the hard recovery guard is inactive. The implementation does not reserve a fixed number of candidates for each subset. Instead, it builds an ordered pool from the base actor, background/safety prior, quality actor, chase action, geometry-aware anchored action when enabled, lateral passing templates, braking templates, recovery templates, and local action perturbations. Duplicate candidates are removed after rounding each action component to three decimals, and the first 12 unique candidates are retained.

The ordered candidate pool starts with the base actor action and the speed-limited anchor action from the barrier expert. If the quality actor is available, its action is appended as an extra target. Under overtake-aware planning, the pool then adds a chase action, optionally a geometry-anchored blend of the base and anchor actions, lateral passing templates, a close-gap braking template, recovery templates when recovery is needed, and local perturbations around the base action.

For passing templates, attack is permitted when recovery is not needed, $|\ell| < 0.34$, heading cosine exceeds 0.84, and a forward or alongside vehicle exists. In sharp-turn geometry, the passing throttle is 0.52; otherwise

it is 0.64. The passing brake is 0. The steering grid is $\{0.08, 0.16, 0.24\}$ in sharp turns and $\{0.12, 0.22, 0.32\}$ otherwise, with heading correction $-0.12e_\psi$. If the nearest forward gap is below 11 m and $|\Delta l| < 8$ m, the braking template uses steering $-0.25\text{sign}(\Delta l)$, throttle 0.04, and brake 0.46. Recovery templates steer against $2e_\psi + 0.80\ell$, use throttle 0.46–0.52 depending on off-track state, and use brake 0 or 0.18 depending on speed. A hard recovery guard bypasses candidate scoring when the vehicle is backward, severely off track, $|\ell| > 0.62$, or heading cosine is below 0.55; in that case a track-follow recovery action is executed after speed limiting.

A. Mask Handling During Imagined Rollout

During real execution, $\bar{o}_{t+1}$ is rebuilt from simulator telemetry by the full dynamic-neighborhood selector and $\text{Pack}(\cdot)$. During imagined rollout, however, the world model operates only on predicted packed observations and does not have access to new physical telemetry. Therefore, the submitted short-horizon rollout treats the predicted mean observation as a continuous packed state and applies a deterministic post-processing operator $\text{PostPack}(\cdot)$ before the next actor query and graph encoding.

Let $\mu_{t+k+1|t}^{o,(n)}$ be the predicted mean of the next packed observation. The post-processed imagined observation is

$$\hat{o}_{t+k+1|t}^{(n)} = \text{PostPack}(\mu_{t+k+1|t}^{o,(n)}, \hat{\mathbf{m}}_{t+k|t}^{0,(n)}). \quad (20)$$

The continuous self and relation physical features are taken from $\mu_{t+k+1|t}^{o,(n)}$. The validity-mask channels are clipped to $[0, 1]$ and used as soft validity weights in the next masked aggregation:

$$\hat{m}_{t+k+1|t}^{0,k,(n)} = \text{clip}(\mu_{t+k+1|t,m(k)}^{o,(n)}, 0, 1). \quad (21)$$

This matches the submitted implementation: during denormalization, physical slot features are denormalized and mask channels are passed through $\text{np.clip}(\text{mask}, 0.0, 1.0)$. The implementation does not copy masks from the previous imagined observation and does not threshold masks into binary values during imagined rollout. Mask handling is therefore deterministic and is not treated as an additional learned discrete sampling process. Physical vehicle identities are not persistently tracked inside imagined rollouts; at the next real simulator step, slot identities are rebuilt from telemetry by the runtime interaction selector.

B. Scoring Terms

Let $\hat{o}_{t+k|t}^{(n)}$ and $\hat{o}_{t+k+1|t}^{(n)}$ be consecutive imagined packed observations for candidate n , and let

$$\Pi = \{\text{ActorMLP}_{\text{base}}, \text{ActorMLP}_{\text{qual}}\}. \quad (22)$$

The progress reward is

$$R_{\text{prog}} = \hat{\eta}_{t+k+1|t} - \hat{\eta}_{t+k|t}. \quad (23)$$

Let $v_c = \min(v_{\text{ref}}, 18)$. The overtaking reward is

$$\begin{aligned} R_{\text{ovt}} = & 1.2\mathbb{I}[g_f^k < \infty, g_f^{k+1} = \infty] + \text{clip}\left(\frac{g_f^k - g_f^{k+1}}{65}, -0.4, 0.8\right) \\ & + 0.35[n_{\text{ahead}}^k - n_{\text{ahead}}^{k+1}]_+ \\ & + 0.03[v_{t+k+1} - v_c]_+. \end{aligned} \quad (24)$$

The lateral-quality cost is

$$\begin{aligned} C_{\text{lat}} = & |\hat{\ell}_{t+k+1|t}| + 0.55|e_{\psi,t+k+1|t}| + 1.8[|\hat{\ell}_{t+k+1|t}| - 0.32]_+ \\ & + 1.2[0.82 - \cos(e_{\psi,t+k+1|t})]_+. \end{aligned} \quad (25)$$

The off-track cost is

$$\begin{aligned} C_{\text{off}} = & 2.5\mathbb{I}[o^{\text{off}} = 1] + 1.2\mathbb{I}[o^{\text{back}} = 1] \\ & + 0.8[|\hat{\ell}_{t+k+1|t}| - 0.46]_+. \end{aligned} \quad (26)$$

The close-gap cost is

$$C_{\text{close}} = \frac{[10 - g_f^{k+1}]_+}{10}. \quad (27)$$

Since $\hat{\chi}_{t+k+1|t}$ is already a sigmoid probability, the risk cost is

$$C_{\text{risk}} = \hat{\chi}_{t+k+1|t}. \quad (28)$$

The uncertainty cost is

$$C_{\text{unc}} = \text{clip}\left(\frac{1}{d_o} \sum_{d=1}^{d_o} \lambda_{t+k+1|t,d}^o + \lambda_{t+k+1|t}^r, -2, 4\right). \quad (29)$$

The imitation cost is

$$C_{\text{imit}} = \min_{\pi \in \Pi} \|\hat{a}_{t+k|t}^{(n)} - \pi(\hat{o}_{t+k|t}^{(n)})\|_2^2. \quad (30)$$

The planner’s forward interaction corridor is $0 < \Delta f < 65$ m and $|\Delta l| < 18$ m in the packed map-relative observation. The alongside corridor is $|\Delta f| \leq 14$ m and $|\Delta l| < 14$ m. The nearest forward gap g_f is the smallest positive Δf in the forward corridor; if no vehicle satisfies the corridor test, $g_f = \infty$. The planner uses the local wrapped map-relative observation supplied by the telemetry extractor.

IV. Evaluation Metrics and Experimental Protocols

A. Metric Definitions and Statistical Summaries

This section defines the primary metrics referenced by the main manuscript. Let $\mathcal{E}$ be the set of evaluated episodes and let $\mathcal{S} \subseteq \mathcal{E}$ be the subset in which an overtaking event is completed by the target vehicle. For episode e , let N_{ovt}^e be the number of completed overtakes, N_{on}^e the number whose event windows have no off-track exposure, and N_{des}^e the number that are on-track, non-reverse, laterally bounded, heading-consistent, and free of the contact proxy. The binary success indicator is $\mathbb{I}_{\text{succ}}^e = \mathbb{I}[N_{\text{ovt}}^e > 0]$, and the success rate is

$$P_{\text{succ}} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbb{I}_{\text{succ}}^e. \quad (31)$$

The episode-level on-track and desirable proportions are

$$p_{\text{on}}^e = \frac{N_{\text{on}}^e}{\max(N_{\text{ovt}}^e, 1)}, \quad p_{\text{des}}^e = \frac{N_{\text{des}}^e}{\max(N_{\text{ovt}}^e, 1)}. \quad (32)$$

Thus, episodes with no completed overtake contribute zero. The reported rates average these episode-level proportions over all cases:

$$P_{\text{on}} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} p_{\text{on}}^e, \quad P_{\text{des}} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} p_{\text{des}}^e. \quad (33)$$

For successful episodes, the completion time is

$$\bar{T}_{\text{comp}} = \frac{1}{|\mathcal{S}|} \sum_{e \in \mathcal{S}} (t_{\text{complete}}^e - t_{\text{start}}^e). \quad (34)$$

Completion time is reported conditionally over successful overtakes because it is undefined when no pass is completed.

For episode e with horizon T_e , target grass exposure is

$$P_{\text{grass}}^e = \frac{1}{T_e} \sum_{t=1}^{T_e} \mathbb{I}[o_{t,\text{grass}}^e = 1], \quad (35)$$

where $o_{t,\text{grass}}^e$ is the target vehicle's simulator grass flag. The E1 grass summary reported in the main manuscript is conditioned on successful episodes:

$$\bar{P}_{\text{grass}|\text{succ}} = \frac{1}{|\mathcal{S}|} \sum_{e \in \mathcal{S}} P_{\text{grass}}^e. \quad (36)$$

It is not interchangeable with an all-episode off-track mean. For the 200-case E1 table, binary success rates use Wilson 95% confidence intervals. Nonzero episode-level desirable proportions and successful-episode continuous metrics use percentile case bootstrap intervals with 4000 resamples and random seed 2026. When all 200 desirable proportions are zero, the empirical all-zero result is reported without treating a degenerate bootstrap interval as population uncertainty. Conditional completion estimates based on fewer than 20 successes are labelled descriptive and highly unstable.

B. Matched Primary-Endpoint Analysis

The analysis designates binary overtaking success as the primary closed-loop endpoint. All algorithms in E1 share the same 200 case identifiers, defined by benchmark, vehicle count, and simulator seed. For comparator b , let Y_e^D and Y_e^b denote the DNQ-DLC and comparator success indicators for matched case e . The reported effect is

$$\hat{\Delta}_{D-b} = \frac{1}{200} \sum_{e=1}^{200} (Y_e^D - Y_e^b). \quad (37)$$

Its 95% interval is the percentile interval from 50,000 bootstrap resamples of matched case rows, using random seed 20260716. Binary symmetry is tested with the two-sided exact McNemar test on the DNQ-DLC-only and comparator-only cells. Holm adjustment covers the six comparators shown in Table SV. Absolute rates use Wilson intervals only to display each method's scale.

The sample size was fixed at 200 matched cases before comparison, and no case was excluded according to its outcome. Precision is reported with the paired confidence interval rather than post hoc observed power. For Rule

Expert, the interval corresponds to a 6.5–21.5-percentage-point increase.

The two vehicle-count strata are sensitivity analyses. In $N = 4$ –6, the paired difference from Rule Expert is 0.050 (95% CI, -0.033 – 0.133 ; exact $p = 0.327$). In $N = 7$ –8, it is 0.275 (95% CI, 0.138 – 0.413 ; exact $p = 0.000472$). The latter is evidence for operation outside the training vehicle-count range tested here, not for arbitrary traffic-scale or opponent-policy generalization.

C. Latency and Randomized Component-Attribution Protocol

The online decision latency is measured around one full call to the policy and reported as the mean and 95th percentile:

$$\bar{L} = \frac{1}{N_L} \sum_{q=1}^{N_L} L_q, \quad L_{95} = \text{Quantile}_{0.95}(\{L_q\}_{q=1}^{N_L}). \quad (38)$$

Experiment 6 uses a matched-seed randomized matrix over Hairpin and S-Curve, $N = 6/8$, and seeds 6101–6108. For each attribution case, the seed randomizes the background start slots, line spacing, and lateral spacing while retaining the target vehicle in the last start slot. The environment exposes all neighbor identities and each policy packs its own $K = 3$ relation slots. The full controller is compared with fixed identity slots, nearest-neighbor slots, no quality proposal, no learned risk/uncertainty cost, no safety-quality costs, no rule anchor, no handcrafted candidates, and no hard recovery. No run is discarded because of its outcome.

The completed matrix contains 288 valid summaries, zero invalid randomization flags, zero configuration-integrity mismatches, and zero failed cases. Each algorithm has 31–32 unique physical metric signatures. Paired bootstrap results show that the full controller improves desirable overtaking by 0.201 [0.066, 0.341] relative to the no-risk/no-uncertainty variant. Relative to removal of all safety-quality terms, it improves desirable behavior by 0.224 [0.042, 0.397] and lowers grass exposure by 0.136 [0.019, 0.260]. Relative to removal of the quality proposal, it lowers grass exposure by 0.183 [0.086, 0.283]. Fixed-identity, nearest-neighbor, and interaction-priority selection have identical success (30/32) and overlapping paired intervals for the other reported endpoints. Removing handcrafted candidates gives 31/32 success and improves all four tabled point estimates in the $N = 6/8$ matrix. A subsequent, separately frozen $N = 10/12$ mixed-traffic stress test reverses the success finding: removal of handcrafted candidates gives 27/32 success, versus 32/32 for the unchanged full controller. Fixed-identity and nearest-neighbor selection also retain 32/32 success; fixed identity has higher desirable behavior, while the full controller has significantly lower grass exposure than nearest neighbor. These results show that selector and candidate effects depend on traffic density.

TABLE SV

Paired details for the E1 overtaking-success endpoint. Difference is DNQ-DLC minus comparator. Exact p is unadjusted; p_{Holm} covers the six comparisons.

Comparator	n	Difference [95% CI]	Both	DNQ only	Comparator only	Neither	Exact p	p_{Holm}
Rule Expert	200	0.140 [0.065, 0.215]	125	46	18	11	0.000617	0.000617
Safety Rule	200	0.155 [0.085, 0.225]	128	43	12	17	0.000033	0.000066
DLC-JTO	200	0.765 [0.700, 0.830]	14	157	4	25	1.89×10^{-41}	9.46×10^{-41}
PPO legacy	200	0.580 [0.505, 0.650]	52	119	3	26	1.14×10^{-31}	4.56×10^{-31}
SAC legacy	200	0.850 [0.800, 0.895]	1	170	0	29	1.34×10^{-51}	8.02×10^{-51}
TD3 legacy	200	0.465 [0.390, 0.540]	73	98	5	24	1.82×10^{-23}	5.45×10^{-23}

TABLE SVI

Detailed secondary outcomes for retrained DLC and direct rule comparators in E1. Success uses Wilson 95% intervals. Desirable overtaking uses all 200 cases. Completion time is conditioned on successful cases, whose count is shown in parentheses. DLC-JT and DLC-JTO completion estimates are descriptive because only 2 and 18 cases succeeded.

Method	Success	Desirable overtaking	Completion time, steps
DLC-IT	0.000 [0.000, 0.019]	0.000	– (0)
DLC-JT	0.010 [0.003, 0.036]	0.000	1583.5 [1079.0, 2088.0] (2)
DLC-JTO	0.090 [0.058, 0.138]	0.000	1346.1 [1220.4, 1465.1] (18)
Rule Expert	0.715 [0.649, 0.773]	0.381 [0.321, 0.442]	126.0 [96.0, 160.0] (143)
Safety Rule	0.700 [0.633, 0.759]	0.379 [0.319, 0.442]	131.1 [94.7, 174.2] (140)
DNQ-DLC	0.855 [0.800, 0.897]	0.420 [0.363, 0.476]	97.8 [81.8, 117.8] (171)

V. Supplementary Videos

The project portal provides the videos associated with the qualitative sequences in the main paper, together with additional fixed-case stress runs. These videos support visual inspection of the simulated maneuvers. They are not used to estimate success rates or component effects, which are calculated from the complete randomized case sets.

VI. Reproducibility and Evidence Limits

The private repository at <https://github.com/chocho2222/DNQ-DLC> contains the DNQ-DLC code, baseline runners, environment specification, self-trained checkpoints, source data, and media site. It will be made public after manuscript submission. A versioned release will then be archived in a DOI-issuing repository.

The evidence is limited to simulation. The experiments do not establish real-vehicle safety, robustness to perception errors, tire-limit validity, or guaranteed safety against adversarial opponents. The weighted decision score is not a formal safety constraint.

DNQ-DLC is a hybrid planner, so performance cannot be assigned only to its learned modules. The legacy E2/E3 sweeps are deterministic checks, not independent multi-seed trials. PPO, SAC, and TD3 are not training-budget-matched primary baselines, and the DLC budget differs from DNQ-DLC. The quality actor also fails both unseen $N = 8$ validation strata. The supported conclusion is therefore limited to improved overtaking completion for the full controller in the evaluated simulator settings.

References

- [1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” arXiv preprint arXiv:1707.06347, 2017.
- [2] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in Proceedings of the International Conference on Machine Learning, 2018, pp. 1861–1870.
- [3] S. Fujimoto, H. van Hoof, and D. Meger, “Addressing function approximation error in actor-critic methods,” in Proceedings of the International Conference on Machine Learning, 2018, pp. 1587–1596.
- [4] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, “Stable-baselines3: Reliable reinforcement learning implementations,” Journal of Machine Learning Research, vol. 22, no. 268, pp. 1–8, 2021.
- [5] W. Swarting, A. Pierson, J. Alonso-Mora, S. Karaman, and D. Rus, “Deep latent competition: Learning to race using visual control policies in latent space,” 2021, arXiv preprint arXiv:2102.09812.